

Demonstration of CMOS Compatible Remote Reading for Magnetic Shift Register Technology toward Next Generation 3D Memory

Michael Quinsat, Yoshihiro Ueda, Naoharu Shimomura, Nobuyuki Umetsu, Yasuaki Ootera, Jun Iwata, Hiroki Tokuhira, Shinji Miyano, Masatoshi Yoshikawa, Tsuyoshi Kondo, Masumi Saitoh, Masaki Kado.

Frontier Technology R&D institute, Kioxia Corporation, Yokohama, 221-0022, Japan, michael.quinsat@kioxia.com

Three-dimensional implementations of magnetic shift registers (SR) require a practical read scheme compatible with dense integration. We propose a 3D SR based on perpendicularly magnetized nanotubes and a remote magnetic tunnel junction (MTJ) reader sensing the stray field of the terminal domain. Device behavior for small tube diameter has been verified by magnetic simulations. To validate the reading concept experimentally, we fabricated 2D SR devices with remote in-plane MTJs on a 130 nm CMOS IC. Using 50 nm MTJs, we demonstrate field-free write/shift/read operation with CMOS driving and sensing circuits. These results support remote MTJ sensing as a practical route toward 3D ultra-high density magnetic memory based on magnetic nanotubes.

Index Terms—Domain wall, Magnetic tunnel junction, shift registers

I. INTRODUCTION

MAGNETIC domain-wall (DW) shift registers (SR) have long been considered as an alternative to conventional solid-state memories because they combine non-volatility with serial access in a compact magnetic nanowire (NW) [1]-[2]. Their possible 3D architectures are particularly attractive for memory density, where magnetic domains are stored along vertical structures and moved by current-induced torques to a read element.

However, the realization of such 3D devices remains limited not only by the integration of write and read elements, but also by the processing of the magnetic storage wire itself. Earlier devices usually relied on direct integration of the NW with the magnetic tunnel junction (MTJ) free layer [3]-[6]. It imposes strong constraints on etching, material selection, driving circuitry, and design constraints on MTJ reader locations.

In this work [7], we propose addressing this issue using a remote reading MTJ that detects the stray field generated by the last stored domain in 3D magnetic nanotube (NT). We already demonstrated current induced DW motion in Co films made by atomic layer deposition [8], and we believe that pre-existing vertical holes can be in principle filled by such conformal deposition technique. The reading principle of the 3D devices concept is experimentally validated in 2D NW integrated on CMOS IC. Our main objective is to demonstrate the reliable field-free write/shift/read operation using standard small MTJs and independent storage element fabrication.

II. 3D SHIFT-REGISTER CONCEPT AND REMOTE READ PRINCIPLE

Fig.1(a) shows the proposed 3D shift-register concept. Information is stored in vertical magnetic nanotubes with perpendicular magnetic anisotropy directed radially along the tube wall. Binary states correspond to inward and outward radial magnetization, separated by ring-shaped DWs. Micromagnetic simulations [9] indicate that current pulses can shift these domain walls along the NT, while periodic diameter modulation defines stable positions and the domain length. In Fig. 1(b), a NT of 50 nm average diameter supports controlled

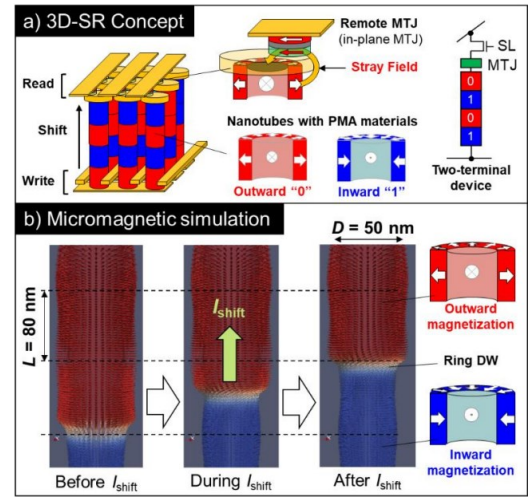


Fig. 1. (a) 3D-SR concept where information is stored as outward or inward magnetization direction in magnetic nanotubes. (b) Micromagnetic simulation evidencing current induce motion of ring DW. From [7]. ©2025 IEEE.

shifting of DW along the NT's length. The read operation is performed at the top of the nanotube by an in-plane MTJ placed above the tube' edge. The last domain generates a stray field that changes sign depending on whether the magnetization is inward or outward. This field biases the MTJ free layer and produces two distinct resistance states. Since NT and MTJ can be serially connected, it can form a 2-terminal device suitable for high density. Fig. 2 shows the simulated stray- field profiles for both NT and NW geometries at the position of the MTJ free layer.

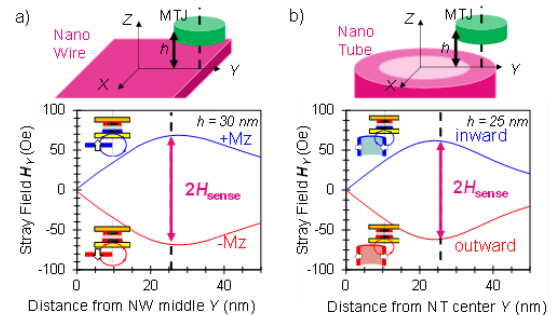


Fig. 2. Stray field vs. the MTJ position Y positioned (a) above a 50-nm wide NW and (b) above a 50-nm diameter NT. From [7]. ©2025 IEEE.

The stray field is strongest just above the edge of the magnetic body in $Y = 25$ nm, leading to a field difference, the read sensitivity $2H_{\text{sense}}$, of about 120 Oe between the 2 different magnetization states. The NW case reproduces well the spatial behavior expected for the NT stray field. This justifies the use of 2D NW devices to validate the remote reading concept experimentally.

III. CMOS-INTEGRATED 2D DEMONSTRATION

We fabricated 2D shift-register arrays on a 130 nm CMOS wafer. Fig. 3 shows the fabricated device with the writing, storing, and reading elements all made in different process steps sequentially. Magnetic domains are written in the nanowire by Oersted fields generated in buried field lines (FL), shifted by shift current pulses in the nanowire, and read by an MTJ.

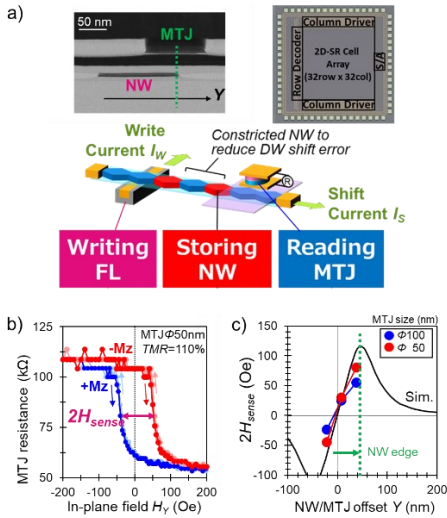


Fig. 3. (a) Cut view of MTJ/NW alignment in fabricated device, chip layout, and 2D-SR device schematic. (b) MTJ resistance vs. in-plane field $+M_z$ or $-M_z$ of the NW. (c) Experimental stray field vs. the MTJ position Y for 50 nm and 100 nm diameter MTJ. Black line is the simulation result. Adapted from [7]. ©2025 IEEE.

The tunnel magnetoresistance ratio (TMR) of the MTJ is above 100% in the fully functional SR devices. The MTJ's switching field depends on the nanowire magnetization state, which directly gives the read sensitivity ($2H_{\text{sense}}$), in Fig.3(b). The measured sensitivity as a function of NW/MTJ offset Y agrees well with the simulated stray-field profile.

In the same chip we made devices with 50 and 100 nm diameter MTJs and a fixed NW shape. Fig.3(c) shows that 50 nm MTJs provide higher sensitivity because the stray field is averaged over a smaller free layer area. This point is important for scaling: reducing the MTJ size compensates for the expected signal reduction from smaller magnetic bits.

Fig. 4 shows the write/shift/read operation using CMOS control circuitry under zero magnetic field. First, a write pulse through the FL creates a domain at the entrance of the NW. Then, shift pulses move the DW toward the remote MTJ. Finally, the MTJ resistance is converted into a digital output by the sensing amplifier. Once a new DW is created, repeated shift pulses displace it under the MTJ, and the output state changes at the expected step in the sequence lasting less than 100 μ s.

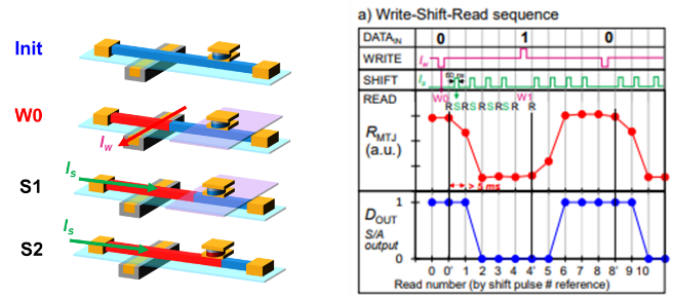


Fig. 4. Field-free I/O operation of 2D-SR integrated on CMOS chip. Adapted from [7]. ©2025 IEEE.

IV. CONCLUSION

We have proposed a magnetic shift-register concept combining 3D magnetic nanotube with remote MTJ sensing, and experimentally validated the read principle in 2D devices integrated on CMOS IC using conventional in-plane MTJs. Because the MTJ and the magnetic storing nanowire are physically decoupled, the approach avoids the main fabrication tradeoff of direct-read structures and leaves freedom for DW control by wire shaping. For 3D memory, this architecture is a promising route toward ultra-high density 3D magnetic memory because the projected cell footprint is determined mainly by the nanotube diameter and access elements.

REFERENCES

- [1] S. S. P. Parkin and S.-H. Yang, "Memory on the racetrack," *Nature Nanotechnology*, vol. 10, pp. 195–198, 2015.
- [2] S. S. P. Parkin, M. Hayashi, and L. Thomas, "Magnetic domain-wall racetrack memory," *Science*, vol. 320, no. 5873, pp. 190–194, 2008.
- [3] A. J. Annunziata, M. C. Gaidis, L. Thomas, C. W. Chien, C. C. Hung, P. Chevalier, E. J. O'Sullivan, J. P. Hummel, E. A. Joseph, Y. Zhu, T. Topuria, E. Delenia, P. M. Rice, S. S. P. Parkin, and W. J. Gallagher, "Racetrack memory cell array with integrated magnetic tunnel junction readout," in 2011 International Electron Devices Meeting, pp. 24.3.1–24.3.4, 2011.
- [4] T. Kondo, H. Morise, T. Shimada, M. Quinsat, M. Kado, Y. Ootera, and S. Nakamura, "A magnetic shift register with periodic potential energy modulation," in 2017 International Symposium on VLSI Technology, Systems and Application, pp. 1–2, 2017.
- [5] E. Raymenants, O. Bultynck, D. Wan, T. Devolder, K. Garello, L. Souriau, A. Thiam, D. Tsvetanova, Y. Canvel, D. E. Nikonov, I. A. Young, M. Heyns, B. Soree, I. Asselberghs, I. Radu, S. Couet, and V. D. Nguyen, "Nanoscale domain wall devices with magnetic tunnel junction read and write," *Nature Electronics*, vol. 4, pp. 392–398, 2021.
- [6] H. Honjo, S. Fukami, T. Suzuki, R. Nabeshi, N. Ishiwata, S. Miura, N. Sakimura, T. Sugibayashi, N. Kasai, and H. Ohno, "Domain-wall-motion cell with perpendicular anisotropy wire and in-plane magnetic tunneling junctions," *Journal of Applied Physics*, vol. 111, 07C903, 2012.
- [7] M. Quinsat, Y. Ueda, N. Shimomura, S. Hashimoto, N. Umetsu, Y. Ootera, J. Iwata, H. Tokuhira, S. Miyano, M. Yoshikawa, T. Kondo, M. Saitoh, and M. Kado, "Demonstration of Reliable Magnetic Shift Register Reading Using 50 nm MTJs on CMOS IC Towards 3D Ultra-High Density Memory," in 2025 9th IEEE Electron Devices Technology & Manufacturing Conference, pp. 1–3, 2025.
- [8] M. Kado, Y. Tokuda, Y. Ootera, N. Umetsu, M. Quinsat, and T. Kondo, "Atomic Layer Deposition of Perpendicularly Magnetized Co Layers Showing Current-Induced Domain Wall Motion," *IEEE Transactions on Magnetics*, vol. 59, no. 11, pp. 1–5, 2023.
- [9] N. Umetsu, H. Tokuhira, M. Quinsat, H. Horii, T. Kondo, and M. Kado, "Spin-orbit torque driven motion of chiral domain walls induced by radial magnetization in nanotube geometry," *Physical Review B*, vol. 112, 094425, 2025.